

π^0 Production Analysis: R_{AA} and R_{CP}

An outline with machine learning hooks, including systematic bands as an ML problem

Working notes

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1 Physics motivation and observables

The nuclear modification factor R_{AA} and the central-to-peripheral ratio R_{CP} are the standard observables for parton energy loss in the QGP. For neutral pions reconstructed via $\pi^0 \rightarrow \gamma\gamma$ (BR $\approx 98.8\%$):

$$R_{AA}(p_T; \text{cent}) = \frac{1}{\langle N_{\text{coll}} \rangle_{\text{cent}}} \frac{(d^2 N^{\pi^0} / dp_T dy)_{\text{cent}}^{AA}}{(d^2 N^{\pi^0} / dp_T dy)^{pp}} \quad (1)$$

$$R_{CP}(p_T; \text{cent}) = \frac{\langle N_{\text{coll}} \rangle_{\text{periph}}}{\langle N_{\text{coll}} \rangle_{\text{cent}}} \frac{(d^2 N^{\pi^0} / dp_T dy)_{\text{cent}}^{AA}}{(d^2 N^{\pi^0} / dp_T dy)_{\text{periph}}^{AA}} \quad (2)$$

Final deliverable: $R_{AA}(p_T)$ and $R_{CP}(p_T)$ curves, one per centrality class (e.g. 0–10%, 10–20%, $\dots$, 60–80%), with statistical errors and correlated systematic uncertainty bands. R_{CP} is the fallback when no clean pp reference exists at the same $\sqrt{s_{NN}}$ – the peripheral spectrum is treated as the proxy reference, but the $\langle N_{\text{coll}} \rangle$ ratio carries large correlated uncertainty so R_{CP} is most useful in shape, less so in normalization.

2 Analysis pipeline

2.1 Event selection and centrality classification

- Minimum bias and high- E_T EMCal triggers; track each trigger class separately and combine downstream with trigger-efficiency weights.
- Quality cuts: $|z_{\text{vtx}}|$, pileup veto, beam-background veto.
- Centrality from V0/T0/ZDC sum or charged-particle multiplicity, mapped to $\langle N_{\text{coll}} \rangle$ and $\langle N_{\text{part}} \rangle$ through a Glauber MC.
- Bin in centrality, then in p_T within each centrality class.

ML hook 0 – centrality calibration. A small regression network mapping (V0 + ZDC + tracker multiplicity, run number) to a calibrated centrality percentile, trained against the Glauber-fit target. Useful where classical NBD fits leak across boundaries, particularly in peripheral bins. Caveat: centrality feeds directly into $\langle N_{\text{coll}} \rangle$, so any ML estimate must reproduce the data-driven Glauber mapping. Cross-validate against the standard table; this is an internal-consistency exercise, not a place to invent new central values.

2.2 Photon reconstruction in the EMCal

- Cluster finder; energy threshold $\sim 0.5\text{--}1$ GeV.
- Cluster-shape cuts (λ_0^2 , ellipticity) to suppress hadronic showers and merged showers.
- Charged-track veto via track-matching ($\Delta\eta$, $\Delta\phi$ to nearest extrapolated track).
- Timing cuts to remove out-of-bunch and pileup clusters.

ML hook 1 – photon ID. BDT or small CNN classifying clusters as photon / merged- π^0 / hadron, using shower-shape moments (λ_0^2 , λ_1^2 , dispersion, N_{cells}), cell-level energies, time, track-matching χ^2 . Train on tagged π^0 in data (sideband-subtracted) plus PYTHIA+GEANT MC. Output is a continuous photon probability, replacing hard cuts. At high p_T a CNN or graph network over the local cell map outperforms scalar BDTs because merged-shower topology becomes the dominant discriminating feature.

2.3 π^0 reconstruction

- Pair all photon candidates per event; compute $M_{\gamma\gamma}$ and $p_T^{\gamma\gamma}$.
- Asymmetry cut $\alpha = |E_1 - E_2|/(E_1 + E_2) < \alpha_{\text{cut}}$ (typically 0.7–0.8).
- Combinatorial background from event mixing or rotation; subtract in $M_{\gamma\gamma}$.
- Per (p_T , centrality) bin: fit signal (Gaussian or Crystal Ball) plus residual background (low-order polynomial) and extract raw yield $N_{\text{raw}}^{\pi^0}(p_T, \text{cent})$.
- At very high p_T the two decay photons merge into a single cluster – switch to a “merged- π^0 tagging” workflow based on cluster shape.

ML hook 2 – signal/background and merged- π^0 .

- Pair-level classifier with features ($M_{\gamma\gamma}$, α , p_T , $\Delta\phi$, cluster qualities) to discriminate signal from combinatorial pairs, replacing the asymmetry/mass-window cuts with a continuous score.
- For merged- π^0 : CNN over a small ($\sim 7 \times 7$) cell window centered on the cluster, outputting the probability of being a merged π^0 . This is the high- p_T workhorse ($p_T \gtrsim 20\text{--}30$ GeV/c).
- Mandatory: *decorrelate* the classifier output from $M_{\gamma\gamma}$ (DisCo penalty, adversarial decorrelation, or planing). Sculpting the background under the peak is the single largest failure mode of ML in spectrum analyses; if the score reshapes the background, the fit-extracted yield is biased and the systematic table cannot recover.

2.4 Efficiency, acceptance, and energy corrections

- PYTHIA / HIJING (or AMPT) event generation embedded into real AA events for occupancy realism, plus GEANT detector simulation.
- Extract $\epsilon(p_T, \text{cent}) = N_{\text{rec}}/N_{\text{gen}}$ in each centrality bin.
- Energy non-linearity correction tuned on test-beam data and in-situ π^0 peak position vs. p_T .
- Conversion correction (photons converting upstream of the calorimeter).
- Trigger efficiency turn-on from MB overlap.

ML hook 3 – response and unfolding.

- *Calibration regression.* Neural network mapping (raw cluster energy, η , ϕ , shower shape, local occupancy) to true photon energy. Trained on single-photon MC, cross-checked against the π^0 mass peak position vs. p_T . Captures centrality-dependent underlying-event leakage into the cluster that a 1D non-linearity correction cannot.
- *ML-based unfolding* (OmniFold-style reweighting, or a conditional normalizing flow) to invert detector response and centrality migration in one step. Strong motivation in HI: the response varies sharply with local occupancy, and a per-bin efficiency factor cannot encode that without re-binning the whole analysis.

2.5 pp reference (for R_{AA})

- Measured pp at the same $\sqrt{s}$, propagated through the same software stack and the same trigger logic.
- If energies differ slightly: $\sqrt{s}$ -scaled interpolation (NLO pQCD or a fit to the world pp data ensemble).

2.6 Constructing R_{AA} and R_{CP}

For each centrality bin i and p_T bin j :

$$R_{AA}^{(i,j)} = \frac{1}{\langle N_{\text{coll}} \rangle_i} \frac{N_{\text{raw}}^{\pi^0}(p_T^j, \text{cent}_i) / (\epsilon_{ij} \text{BR} \Delta p_T^j \Delta y N_{\text{evt}}^i)}{(d^2 N / dp_T dy)_j^{pp}}.$$

R_{CP} uses the same numerator construction, with the peripheral bin's corrected spectrum (rescaled by the $\langle N_{\text{coll}} \rangle$ ratio) as the denominator. The output is a 2D table: R_{AA} and R_{CP} vs. p_T , one curve per centrality class.

3 Systematic uncertainties: the conventional recipe

Per (p_T , centrality) bin, the conventional contributions are:

- Cluster-shape and timing cut variation
- Track-matching / conversion correction
- Asymmetry cut variation
- Mass-window and background-fit-model variation
- EMCal energy scale and non-linearity
- Trigger efficiency and turn-on shape
- Material budget
- Glauber: $\langle N_{\text{coll}} \rangle$ (dominates correlated systematic, especially peripheral)
- Unfolding regularization, if used

The conventional recipe is: vary each cut by $\pm\delta$ in a few discrete steps, repeat the analysis, take the spread (often with a Barlow-significance check), combine in quadrature, propagate to R_{AA} . Bin-to-bin and source-to-source correlations are handled by hand via covariance matrices.

This recipe has well-known weaknesses, and they map naturally onto an ML problem:

- The size of each variation δ is chosen by the analyst.
- The variations are discrete; their spread is reported as if Gaussian.
- Correlations across cut sources are usually ignored or assumed diagonal.
- Cuts that interact non-linearly (shower shape $\times$ asymmetry, for instance) are not captured by independent 1D scans.
- The final band has no calibrated probabilistic interpretation.

4 Systematic band quantification as an ML problem

This is the place I would actually push ML hardest, and the framing I would advocate. Stop treating the systematic as a discrete cut-variation table and start treating it as posterior inference over nuisance parameters.

4.1 Framing

Let

$$\theta = (\theta_{\text{cluster}}, \theta_{\text{asym}}, \theta_{\text{mass}}, \theta_{\text{scale}}, \theta_{\text{Ncoll}}, \theta_{\text{material}}, \theta_{\text{ID-NN}}, \dots)$$

be the vector of nuisance parameters – every cut threshold, every calibration knob, every model choice. The analysis is then a deterministic (or noisy) function

$$R_{AA}(p_T; \text{cent}; \theta).$$

The systematic band is the spread of this function under a prior $\pi(\theta)$ that encodes what we believe about the nuisance parameters. Conventional analyses sample θ at a handful of corner points and call the spread the systematic. The ML framing is: learn $\theta \mapsto R_{AA}(p_T, \text{cent}; \theta)$ as a surrogate and integrate it against $\pi(\theta)$. Everything else is implementation detail.

4.2 Concrete approaches, ordered by cost to start

(a) Surrogate model of the analysis. Run the full analysis at ~ 200 – 1000 quasi-random Latin-hypercube points in θ -space (parallelizable, embarrassingly so). For each point, store the resulting vector $\{R_{AA}(p_T^j, \text{cent}_i)\}$. Fit a Gaussian process (or a small multi-output neural net) over the whole grid so that bin-to-bin correlations are learned jointly. The systematic band is

$$\sigma_{\text{syst}}^2(p_T^j, \text{cent}_i) = \text{Var}_{\theta \sim \pi(\theta)} \left[\hat{R}_{AA}^{\text{surrogate}}(p_T^j, \text{cent}_i; \theta) \right]$$

and the bin-to-bin covariance matrix falls out automatically. This is the single highest-value upgrade over the cut-variation table: it gives a calibrated covariance instead of a list of numbers, and it makes the prior explicit.

(b) Bayesian or ensembled ML classifiers. Wherever an ML classifier appears in the chain (photon ID, signal/background, merged- π^0), propagate its own epistemic uncertainty:

- Ensemble of $N \sim 20$ classifiers across different seeds, architectures, and training samples.
- MC-dropout at inference for a Bayesian-NN approximation.
- Deep ensembles with last-layer Laplace for a cheap predictive covariance.

The spread of R_{AA} over the ensemble is the classifier-induced systematic. Fold it into the surrogate above as just another nuisance dimension.

(c) Decorrelation / pivotal training. Train classifiers with an adversarial head, a DisCo penalty, or a planing transform so that the score is independent of $M_{\gamma\gamma}$ and of the variables (run period, occupancy, data-vs-MC indicator) that drive the dominant systematics. This shrinks the systematic at its source instead of estimating it after the fact. The reduction is measurable: report the residual mutual information between the score and the protected variable.

(d) Domain adaptation between MC and data. MC/data closure is one of the worst-controlled systematics in HI. Train a domain-adversarial classifier so that intermediate features cannot distinguish MC from data. The residual data-MC disagreement after adaptation is the systematic, and the Wasserstein distance between adapted MC and data control samples gives a quantitative handle.

(e) Simulation-based inference (SBI) on the full likelihood. The cleanest framing. Treat R_{AA} as a parameter of interest jointly with θ . Train a normalizing flow $p(R_{AA} | \text{summary statistics}, \theta)$ on a large MC bank generated with θ sampled from its prior. At analysis time, condition on the observed summary statistics and read off the posterior on R_{AA} marginalized over θ . The credible interval *is* the systematic band. Expensive but principled; this is where HEP is converging for unfolded measurements.

(f) Generative models for detector-response uncertainty. Train a conditional normalizing flow or VAE on a data control sample (isolated photons, identified electrons) to model the residual mismatch between simulated and observed shower response. Sample from this generative model to inject extra variations into MC, then propagate through the surrogate. This gives the “non-linearity” systematic an actual probabilistic shape, instead of a flat $\pm 1\%$ band drawn by hand.

4.3 What the band actually looks like

With the surrogate approach (the realistic first target), the published $R_{AA}(p_T)$ band gains three properties the cut-variation table cannot deliver:

- A bin-by-bin covariance matrix, reusable by theory comparisons and global fits.
- A decomposition into named nuisance directions (Glauber, energy scale, photon ID, ...) via Sobol indices or local sensitivity from the surrogate.
- A defensible probabilistic meaning: a 68% credible region under an explicit prior, rather than a quadrature sum of unrelated variations.

The honest cost: $\pi(\theta)$ is now in the spotlight, and the analyst has to defend it. That is a feature, not a bug, but reviewers will push on it. Plan to ship sensitivity plots – how does R_{AA} shift under “Glauber prior widened by $2\times$ ”, “photon-ID prior shifted by one ensemble member”, etc.

5 Closure tests and blinding

- MC closure: inject a known suppression, recover it through the full pipeline; expect agreement within statistical uncertainty.
- Bottom-line closure: compare the ML pipeline against the classical cut-based pipeline on a fraction of unblinded data. Any difference becomes a “method” systematic until resolved.
- Keep the high- p_T region blinded until calibration, efficiency, and the systematic surrogate are all frozen.

6 Where I would actually start

1. Build the classical, cut-based pipeline end-to-end with conservative cuts. Get a usable R_{AA} curve out before touching ML at all.
2. Add the photon-ID BDT/CNN (ML hook 1) as a drop-in replacement for the cluster-shape cut. Compare against the cut-based result. If they disagree, that disagreement is the first calibration target, not the headline.
3. Build the surrogate-model systematic from Section 5.2(a). Even with the classical pipeline as the underlying analysis, this immediately upgrades the systematic table to a covariance matrix.
4. Add classifier ensembling (Section 5.2(b)) for the photon-ID uncertainty contribution.
5. Only after the above closes in MC, attempt the merged- π^0 CNN for the high- p_T extension, and SBI (Section 5.2(e)) for the unfolded result.

The ordering matters. ML in HEP fails most often not because the model is wrong but because the surrounding pipeline has hidden assumptions that the model silently exploits. Build the classical baseline first; let the ML earn its place.